



SEMESTER END EXAMINATIONS – MAY 2023

Program : **B.E. – Information Science and Engineering**
Course Name : **Artificial Intelligence**
Course Code : **ISE644**

Semester : **VI**
Max. Marks : **100**
Duration : **3 Hrs**

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Discuss the Utility based agents with a schematic diagram. CO1 (06)
 - Identify the limitation of minimax algorithm. Write Alpha-Beta pruning algorithm and explain how it overcomes this limitation by solving the game tree in Fig.1(b). Show all the steps. CO1 (08)

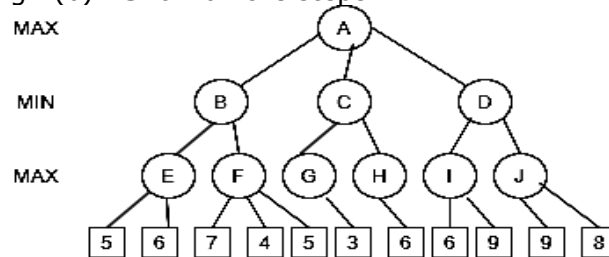


Fig.1(b). : Game tree

- Differentiate the following dimensions to analyze several task environment with examples: CO1 (06)
 - Fully observable vs partially observable
 - Deterministic vs Stochastic
 - Episodic vs sequential.
- With a neat schematic diagram and pseudocode explain the working principles of model-based reflex agent. CO1 (10)
 - Explain with an example A* search and discuss its optimality. CO1 (10)

UNIT - II

- Write truth-table enumeration algorithm for deciding propositional entailment. CO2 (08)
 - Interpret the following propositional logic written for wumpus world KB. CO2 (05)

$$L_{1,1}^0 \wedge FacingEast^0 \wedge Forward^0 \Rightarrow (L_{2,1}^1 \wedge \neg L_{1,1}^1)$$
 - Assuming predicates Parent (p, q) and Female (p) and constants Joan and Kevin, with the obvious meanings, express each of the following sentences in first-order logic. (You may use the abbreviation $\exists 1$ to mean "there exists exactly one."): CO2 (07)
 - Joan has a daughter (possibly more than one, and possibly sons as well)
 - Joan has exactly one daughter (but may have sons as well).
 - Joan has at least one child with Kevin, and no children with anyone else.
- Write and interpret the propositional logic to represent the knowledge derived by each of the following sentences for Wumpus world problem: CO2 (06)
 - Square (2, 2) is breezy.
 - No stench in square (2, 1).
 - Glitter and stench in square (3, 3).

- b) Describe three improvements that DPLL algorithm embodies compared to normal backtracking algorithm. CO2 (06)
- c) Discuss briefly the knowledge engineering process in first order logic. CO2 (08)

UNIT - III

5. a) Propositionalization Approach is rather inefficient. Justify with an example. Explain how unification and Lifting can help resolve the issue of inefficient inferencing. CO3 (05)
- b) Use forward chaining to prove "West is a criminal". The following sentences are stored in the KB.
 --American(x) Weapon(y) sells (x,y,z) hostile(z) Criminal(x)
 --Owns(nono,M1)
 --Missile(M1)
 --American(west)
 --Enemy(x, American) hostile(x)
 --Enemy(nono, American)
 --Missile(x) owns(nono,x) sells(west,a,nono)
 --Missile(x) weapon(x)
 Illustrate with proof tree also. CO3 (05)
- c) Explain with pseudocode, how does the GRAPHPLAN algorithm extract a plan directly from the planning graph. CO3 (10)
6. a) State the differences between the forward and backward chaining with examples. CO3 (08)
- b) Discuss how planning graphs can be used for heuristic estimation with an example. CO3 (06)
- c) Explain how resolution can be extended to first order logic using examples. CO3 (06)

UNIT- IV

7. a) Explain baye's rule and its use with an example. CO4 (08)
- b) Define uncertainty. Explain with an example the meaning of 'agent acting under uncertainty'. CO4 (07)
- c) Discuss any two applications of reinforcement learning. CO4 (05)
8. a) Calculate $P(\text{cavity} \vee \text{toothache})$ by applying the full joint distribution given below: CO4 (04)

	<i>toothache</i>		$\neg$ <i>toothache</i>	
	<i>catch</i>	$\neg$ <i>catch</i>	<i>catch</i>	$\neg$ <i>catch</i>
<i>cavity</i>	0.108	0.012	0.072	0.008
$\neg$ <i>cavity</i>	0.016	0.064	0.144	0.576

- b) Discuss four major factors which are used to improve agent component's learning capability from data. CO4 (08)
- c) Explain with an example active reinforcement learning. CO4 (08)

UNIT - V

9. a) Describe the various agent components that must be viewed while designing AI agents. CO5 (10)
- b) Explain the Weak AI hypothesis in detail. CO5 (10)
10. a) Machines can actually think! In this regard, explain the strong AI hypothesis. CO5 (10)
- b) Write short notes on : CO5 (10)
- Decision-Theoretic Metareasoning
 - Anytime Algorithms
 - Reflective Architecture
